Banach-Tarski and the Paradox of Infinite Cloning

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1 Introduction

What do you think of when you hear the word "infinity"? The symbol ∞ or something that never ends likely come to mind. By definition, infinity is a value greater than any assignable quantity or countable number. In close relation to infinity is cloning, or producing an identical replica of the individual organism or object. Now, consider the following situation: Imagine you and your friend become hungry during a hike, so you decide to split an apple into two. At this point, you think your share of the apple is $\frac{1}{2}$ of the original, but your friend disagrees. She says that theoretically, both pieces of the apples are exactly the same size as the original full-sized apple. What does she mean by this?

This scenario represents the argument called the Paradox of Infinite Cloning, an idea promulgated by mathematicians Stefan Banach and Alfred Tarski in 1924. In essence, this paradox claims that concept of infinite points makes it possible to turn one whole apple into two whole apples. The paradox itself proves that according to the fundamental rules of mathematics, a solid three-dimensional ball can be split into parts that can combine again to form two exact identical copies of the original. This paradox arose largely due to the complex concept of infinity as mentioned earlier. People often mistake infinity to be a real number, and as a result get confused by its unique properties. For instance, you can add or subtract finite numbers to infinity, and the outcome will always be infinity. The difference between a countable and uncountable infinity makes the natural numbers a smaller infinity compared to the real numbers, and this is called a difference in cardinality. In 1891, Georg Cantor proved that the infinite number of points on a line has the same cardinality as the infinite number of points for a volume of a shape.

Cantor's discovery inspired Banach and Tarski's thinking when they realized that they could use an uncountable infinite set of points to turn one sphere into two, or in other words, perform infinite cloning. The Banach-Tarski Paradox is a theorem which states: Given a solid ball in 3D space, there exists a decomposition of the ball into a finite number of disjoint subsets, which can then be put back together in a different way to yield two identical copies of the original ball. The theorem is a paradox because it contradicts the basic principles of geometry. The idea that an object can be doubled by dividing and transforming it without adding any new points seems impossible and supernatural.

Fun Fact: A stronger form of the theorem implies that given any two reasonable solid objects, the cut pieces of either one can be reassembled into the other. This is often stated informally as "a pea can be chopped up and reassembled into the Sun" and called the "pea and the Sun paradox."

2 A Four-Step Proof

There are four basic steps to prove this theorem.

2.1 Step 1

The first step is to find a paradoxical decomposition of the free group in two generators. The free group with two generators a and b consists of all finite strings that can be formed from the four symbols a, a^{-1}, b, and b^{-1} such that no a appears directly next to a^{-1} and no b appears directly next to b^{-1}. The group F_2 is a group that is "paradoxically decomposed," and let $S(a)$ be the set of all non-forbidden strings that start with a. The resulting equation would be:

$$F_2 = e \cup S(a) \cup S(a^{-1}) \cup S(b) \cup S(b^{-1}) = F_2 = aS(a^{-1}) \cup S(a).$$

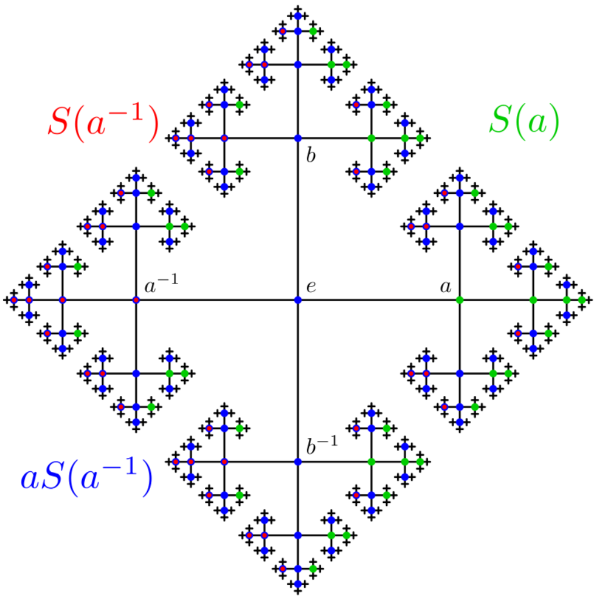


Figure 1: Cayley Graph of F_2

2.2 Step 2

The second step is to find a group of rotations in a 3D space isomorphic to the free group in two generators. In order to find a free group of rotations of 3D space that behaves just like the free group F_2, two orthogonal axes are taken. Another arithmetic proof of the existence of free groups in some special orthogonal groups using integral quaternions lead to paradoxical decomposition of the rotation group.

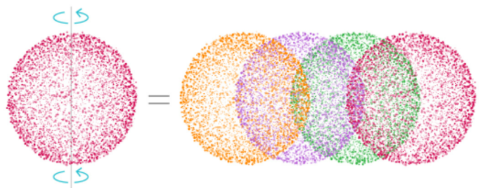
2.3 Step 3

The third step is to use the paradoxical decomposition of that group and the axiom of choice to produce a paradoxical decomposition of the hollow unit sphere. The unit sphere S^2 is separated into orbits by the action of our group H: two points belong to the same orbit if there is a rotation in H which moves the first point into the second. The paradoxical decomposition of H yields a paradoxical decomposition of S^2 into four pieces: A_1, A_2, A_3, A_4.

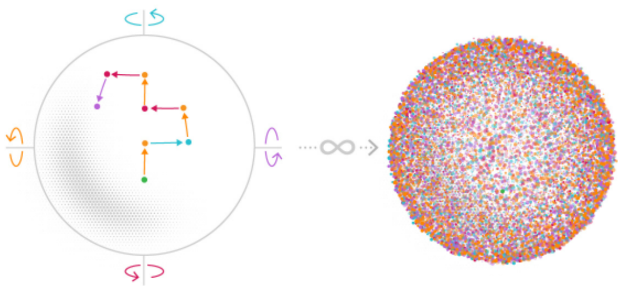
$$A_1 = S(a)M \cup M \cup B$$
$$A_2 = S(a^{-1})M/B$$
$$A_3 = S(b)M$$
$$A_4 = S(b^{-1})M$$

2.4 Step 4

The final step of the proof is to extend this decomposition of the sphere to a decomposition of the solid unit ball. As a result, this connects every point on S^2 with a half-open segment to the origin; the paradoxical decomposition of S^2 then yields the difference between paradoxical decomposition of the solid unit ball and the point at the ball's center. To apply this problem, for example, think of the group that contains all points derived from a final rotation to the East. Rotate this group to the West, and this instantly negates all those final East rotations, transforming the group into the collection of points that immediately preceded the formation of the original sets. The group now contains points that finished on North, South, and East rotations. In other words, the rotated piece contains both the new content and its old self. The nature of infinity makes this seeming increase possible.



Rotating the **East group** to the **West** duplicates the **South**, the **North**, the **starting points**, and the **East group** itself.



3 The Von Neumann Paradox

The Banach-Tarski paradox also corresponds to the Von Neumann paradox, which explains the impossibility of a distinction between planar and high-dimensional cases of the decomposition of a disk of Banach-Tarski using Euclidean congruence. Jon Von Neumann's paradox includes the idea that, unlike the 3D rotation group, the $E(2)$ group of Euclidean motions is indeed solvable, so it rules out the paradoxical decomposition of non-negligible sets. Von Neumann goes on to question, "Can such a paradoxical decomposition be constructed if one allows a larger group of equivalences?" Since any two squares in a plane can become equivalent, a paradoxical decomposition of a square could be counter-intuitive for the same reasoning as the Banach-Tarski decomposition of a ball.

Using the Banach-Tarski paradox can actually strengthen the paradox for a square by claiming that any two bounded subsets of the Euclidean plane with non-empty interiors are equidecomposable with respect to the area-preserving affine maps.

The Von Neumann paradox has made progress in recent years. For example, in 2017, it was found that although the paradox concerns the Euclidean plane, there are other classical groups where paradoxes are possible; in 2019, the paradox was found to use finitely many pieces in the duplication. In return, the class of groups that was isolated by Von Neumann in the study of the Banach-Tarski paradox has been helpful in other areas of mathematics.

4 Conclusion

Today, mathematicians consider the Banach-Tarski paradox as a demonstration of the richness of mathematics. It offers a law-abiding example of how mathematics can stray from physical intuition without contradicting itself. As American mathematician Philip J. Davis once reflected, "One of the endlessly alluring aspects of mathematics is that its thorniest paradoxes have a way of blooming into beautiful theories."

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